

# Student Reading Preference Classification Using PCA-Based Clustering and Collaborative Filtering for Personalized Book Recommendations

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**Abstract**—School libraries face the challenge of helping students discover books that match their diverse reading preferences. This paper presents a hybrid approach combining Principal Component Analysis (PCA)-based clustering for student classification and user-based collaborative filtering for personalized recommendations. Using a dataset of 100 students rating 100 books (40% sparsity), we first apply PCA for dimensionality reduction, revealing that PC1 (28.2% variance) represents STEM vs ARTS preferences while PC2 (18.7% variance) captures reading intensity. K-means clustering on the principal components identifies three distinct student segments. Genre-based classification further refines students into STEM (68%) and ARTS (32%) categories. The collaborative filtering component, using Pearson correlation for user similarity, generates personalized Top-2 recommendations with predicted ratings between 4.03-4.94. Our results demonstrate that PCA-based classification provides interpretable student segments that can guide both collection development and personalized recommendation strategies. The hybrid model achieves strong cluster separation (Silhouette score = 0.37 for k=2) while maintaining recommendation quality, offering practical insights for library management and educational planning.

**Keywords:** PCA, clustering, student classification, collaborative filtering, recommendation system, STEM vs ARTS, library science

## I. INTRODUCTION

### A. Research Context

The explosion of available information in modern libraries has created a paradox: while students have access to more books than ever, finding content that matches their individual preferences becomes increasingly challenging. Traditional library browsing and manual recommendations by librarians, while valuable, do not scale effectively for large collections and diverse student populations [1].

Recommendation systems have proven transformative in commercial settings such as e-commerce and streaming platforms, where they guide users to relevant content based

on behavioral data [2]. However, in educational contexts like school libraries, recommendations serve a dual purpose: not only helping students find books they will enjoy, but also understanding aggregate reading preference patterns to inform collection development and curriculum design.

This research addresses a critical gap: while collaborative filtering techniques effectively generate personalized recommendations, they do not inherently provide insights into student segmentation. Conversely, clustering methods reveal group structures but may not capture the nuanced similarities needed for individual recommendations. Our hybrid approach leverages the strengths of both paradigms.

### B. Problem Statement

School libraries possess valuable rating data from students for books they have read, typically on a 1-5 scale (1 = strongly dislike, 5 = strongly like). This data presents two key opportunities:

- 1) **Classification Challenge:** Can we identify meaningful student segments based on reading preferences? Specifically, can we distinguish between students with STEM (Science, Technology, Engineering, Mathematics) versus ARTS (novels, literature, creative content) orientations?
- 2) **Recommendation Challenge:** Given sparse rating data (60% of possible student-book pairs have ratings), can we accurately predict ratings for unread books and generate personalized recommendations?

Traditional approaches treat these as separate problems. We hypothesize that an integrated solution provides superior insights: PCA-based clustering for interpretable segmentation, followed by collaborative filtering within and across segments for recommendations.

### C. Research Contributions

This work makes several contributions to recommender systems in educational contexts:

- 1) **Methodological Integration:** We demonstrate how PCA can serve dual purposes - dimension reduction for efficient clustering and interpretable feature extraction (PC1 as STEM/ARTS axis).
- 2) **Student Segmentation:** Using 100 students and 100 books, we identify three robust clusters via K-means on principal components, with further refinement into STEM/ARTS categories showing 68% STEM preference among students.
- 3) **Clustering Validation:** We employ multiple validation metrics (Elbow Method, Silhouette Score, Gap Statistic) and show that  $k=2$  or  $k=3$  provides optimal balance between cluster separation and interpretability.
- 4) **Practical Implications:** Our analysis reveals that genre correlations (e.g., Technology-Science positive, Manga-History negative) validate the STEM/ARTS dichotomy and can guide collection acquisition strategies.
- 5) **Recommendation Quality:** Despite 40% sparsity, our collaborative filtering model achieves high predicted ratings (4.03-4.94) for Top-2 recommendations across five test students, demonstrating practical utility.

### D. Paper Organization

The remainder of this paper is organized as follows: Section II reviews related work in recommender systems, PCA, and clustering. Section III details our methodology including data preprocessing, PCA application, K-means clustering, and collaborative filtering. Section IV presents comprehensive results emphasizing PCA-based classification outcomes. Section V discusses interpretation and practical implications. Section VI concludes with limitations and future work.

## II. RELATED WORK

### A. Collaborative Filtering

Collaborative filtering (CF) remains one of the most successful approaches to recommendation systems. User-based CF, pioneered by GroupLens [7], identifies users with similar preferences and recommends items liked by similar users. Pearson correlation is widely used to measure user similarity due to its mean-centering property, which accounts for varying rating tendencies [8].

Matrix factorization techniques, particularly Singular Value Decomposition (SVD), have shown superior performance in handling sparsity by discovering latent features [3]. However, these methods sacrifice interpretability. For educational applications where understanding *why* recommendations work matters, we prioritize transparent similarity-based approaches.

### B. Principal Component Analysis

PCA, introduced by Pearson and Hotelling, transforms high-dimensional data into orthogonal components capturing maximum variance [4]. In recommendation systems, PCA primarily serves as preprocessing for dimensionality reduction. However, principal components themselves can be highly interpretable.

Recent work has shown that in rating matrices, PC1 often captures a "general preference" dimension while subsequent components reveal genre-specific or demographic patterns. We exploit this property to interpret PC1 as a STEM/ARTS preference axis.

### C. Clustering for User Segmentation

K-means clustering groups users into segments for targeted marketing or content curation [5]. Determining optimal  $k$  remains challenging; the Elbow Method, Silhouette Score, and Gap Statistic [6] provide complementary perspectives.

In educational analytics, clustering has been applied to learning styles and performance patterns. Our work extends this to reading preferences, providing a foundation for personalized learning resource recommendations.

### D. Hybrid Approaches

Hybrid recommender systems combine multiple techniques to leverage their strengths. Common combinations include CF with content-based filtering, or CF with demographic information. Our hybrid approach is novel in educational contexts: PCA-based classification to understand student segments, then collaborative filtering for personalized recommendations informed by segment membership.

## III. METHODOLOGY

### A. Model Assumptions

This model is built upon eight core assumptions that underpin both the collaborative filtering and clustering approaches:

- 1) **Preference Similarity:** Students with similar rating patterns share the same reading tastes. A high correlation indicates aligned preferences — this is the fundamental assumption of collaborative filtering.
- 2) **Rating Consistency:** Ratings given by students are consistent and stable within the observation period. Preferences do not change drastically over time.
- 3) **Missing At Random (MAR):** Empty values are caused by students not having read the book yet, not because they are actively avoiding books they dislike.
- 4) **Rating Honesty:** Students provide ratings that genuinely reflect their preferences, without external bias such as peer pressure or grade considerations.
- 5) **Preference Linearity:** Rating deviations from the mean can be linearly combined for prediction, following the principle of weighted average in collaborative filtering.
- 6) **Interval Rating Scale:** Differences between rating values are treated as equal; the gap between ratings

1–2 is the same as between ratings 4–5 (treating ordinal data as interval).

- 7) **Individual Rating Baseline:** Each student has a different baseline tendency in giving ratings (some are "easy raters", others are "hard raters"), accommodated via mean-centering in Pearson correlation.
- 8) **Genre Representation:** Book genres can represent STEM and ARTS categories as a proxy for classifying reading preferences. We assume genre labels are accurate and meaningful to students.

These assumptions are common in recommendation system literature but should be validated in practice. Violations may impact model performance; we address some limitations in Section V.

### B. Dataset Description

Our dataset comprises ratings from 100 students across 100 books in a school library collection. Key characteristics:

- **Rating Scale:** 1-5 (1 = strongly dislike, 5 = strongly like)
- **Total Ratings:** 6,000 entries
- **Sparsity:** 40% (meaning 60% of possible student-book pairs have ratings)
- **Average Ratings per Student:** 60 books
- **Average Ratings per Book:** 60 students
- **Genres:** 12 categories including Novel, Technology Basics, Non-fiction Education, Science Popular, Biography, Manga, History, Literature, Skills, etc.

Books are labeled BXXX (B001-B100) and students SXXX (S001-S100). Genre labels enable interpretation of clusters as STEM-oriented versus ARTS-oriented.

### C. Data Preprocessing

1) *Handling Missing Values:* The rating matrix  $\mathbf{R} \in \mathbb{R}^{100 \times 100}$  has 40% missing entries. For collaborative filtering, we preserve these as missing (to be predicted). For PCA and clustering, we employ mean imputation: missing ratings for book  $j$  are replaced with  $\bar{r}_j = \frac{1}{|U_j|} \sum_{u \in U_j} r_{u,j}$ , where  $U_j$  is the set of students who rated book  $j$ .

Mean imputation assumes Missing At Random (MAR): students have not read the book yet, rather than actively avoiding disliked books.

2) *Standardization:* Before PCA, we standardize the rating matrix to zero mean and unit variance per feature (book):

$$z_{i,j} = \frac{r_{i,j} - \bar{r}_j}{\sigma_j} \quad (1)$$

where  $\bar{r}_j$  and  $\sigma_j$  are the mean and standard deviation of ratings for book  $j$ . This ensures all books contribute equally to principal components regardless of rating scale variations.

### D. Principal Component Analysis

1) *PCA Computation:* Given the standardized matrix  $\mathbf{Z} \in \mathbb{R}^{100 \times 100}$ , we compute the covariance matrix:

$$\mathbf{C} = \frac{1}{n-1} \mathbf{Z}^T \mathbf{Z} \quad (2)$$

Eigenvalue decomposition yields:

$$\mathbf{C} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^T \quad (3)$$

where  $\mathbf{W}$  contains eigenvectors (principal directions) and  $\mathbf{\Lambda}$  is a diagonal matrix of eigenvalues (variance explained by each component).

Student projections onto principal components:

$$\mathbf{Y} = \mathbf{Z} \mathbf{W} \quad (4)$$

2) *Component Selection:* We retain components explaining  $> 70\%$  cumulative variance. The scree plot visualizes variance explained versus component number. In our case, PC1-PC3 explain 53.9% variance, with PC1 and PC2 capturing 46.9%.

3) *Interpretation of Principal Components:* **PC1 (28.2% variance):** Analysis of loading weights reveals PC1 positively correlates with STEM genres (Technology Basics, Science Popular, Non-fiction Education) and negatively with ARTS genres (Novel, Literature). Thus, PC1 represents a *STEM vs ARTS preference axis*.

**PC2 (18.7% variance):** PC2 captures *reading intensity and secondary preferences*. High PC2 values correspond to students who rate many books highly, with variance in non-fiction sub-genres (History vs Education).

### E. K-Means Clustering

1) *Clustering on Principal Components:* We apply K-means on  $\mathbf{Y}[:, 1 : 2]$  (PC1 and PC2 only) to reduce noise from higher-order components. K-means minimizes:

$$J = \sum_{k=1}^K \sum_{\mathbf{y} \in C_k} \|\mathbf{y} - \boldsymbol{\mu}_k\|^2 \quad (5)$$

where  $C_k$  is the set of students in cluster  $k$  and  $\boldsymbol{\mu}_k$  is the cluster centroid.

Algorithm:

- 1) Initialize  $K$  centroids randomly
- 2) Assign each student to the nearest centroid
- 3) Update centroids:  $\boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{\mathbf{y} \in C_k} \mathbf{y}$
- 4) Repeat steps 2-3 until convergence

We run K-means with 25 random initializations to avoid local minima.

2) *Determining Optimal K:* **Elbow Method:** Plot within-cluster sum of squares (WCSS) vs  $K$ . The "elbow" point where WCSS reduction plateaus suggests optimal  $K$ . Our elbow appears at  $K=2$  or  $K=3$ .

**Silhouette Score:** Measures how similar a student is to their own cluster versus other clusters:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (6)$$

where  $a(i)$  is the mean distance to other students in the same cluster, and  $b(i)$  is the mean distance to students in the nearest different cluster. Range:  $[-1, 1]$ ; higher is better. We achieve maximum Silhouette score ( $\approx 0.37$ ) at  $K=2$ .

**Gap Statistic:** Compares total WCSS to its expected value under a null reference distribution (uniformly distributed

data). The optimal  $K$  maximizes Gap statistic. Our Gap statistic is relatively stable for  $K=2$  to  $K=10$ , not providing a clear optimum.

**Decision:** Based on Silhouette score and interpretability, we present results for  $K=3$  (three distinct reading behavior groups) and further simplification to  $K=2$  via genre-based STEM/ARTS classification.

#### F. Genre-Based STEM/ARTS Classification

To enhance interpretability, we classify students into STEM, ARTS, or Balanced (BAL) categories based on genre preferences:

- 1) Compute mean rating per genre for each student using predicted complete matrix
- 2) Define STEM genres: Technology Basics, Non-fiction Science, Non-fiction Education, Science Popular
- 3) Define ARTS genres: Novel, Young Adult Novel, Literature
- 4) Calculate:

$$\text{STEM\_score} = \text{mean}(\text{ratings on STEM genres}) \quad (7)$$

$$\text{ARTS\_score} = \text{mean}(\text{ratings on ARTS genres}) \quad (8)$$

- 5) Classify:

- STEM: if  $\text{STEM\_score} > \text{ARTS\_score} + 0.3$
- ARTS: if  $\text{ARTS\_score} > \text{STEM\_score} + 0.3$
- BAL: if  $|\text{STEM\_score} - \text{ARTS\_score}| \leq 0.3$

The threshold 0.3 is chosen heuristically; future work should optimize this via cross-validation.

#### G. Collaborative Filtering for Recommendations

1) *User Similarity via Pearson Correlation:* For students  $u$  and  $v$ , Pearson correlation on co-rated books:

$$\text{sim}(u, v) = \frac{\sum_{i \in I_{uv}} (r_{u,i} - \bar{r}_u)(r_{v,i} - \bar{r}_v)}{\sqrt{\sum_{i \in I_{uv}} (r_{u,i} - \bar{r}_u)^2} \sqrt{\sum_{i \in I_{uv}} (r_{v,i} - \bar{r}_v)^2}} \quad (9)$$

where  $I_{uv}$  is the set of books rated by both students. We compute a full  $100 \times 100$  similarity matrix using pairwise complete observations.

2) *Rating Prediction:* To predict student  $u$ 's rating for book  $i$ :

$$\hat{r}_{u,i} = \bar{r}_u + \frac{\sum_{v \in N_k(u,i)} \text{sim}(u, v) \cdot (r_{v,i} - \bar{r}_v)}{\sum_{v \in N_k(u,i)} |\text{sim}(u, v)|} \quad (10)$$

where  $N_k(u, i)$  is the set of top- $k$  most similar students to  $u$  who have rated book  $i$ . We use  $k = 10$ .

Predictions are clipped to  $[1, 5]$  to maintain valid rating range.

3) *Generating Recommendations:* For each target student:

- 1) Identify books not yet rated
- 2) Predict ratings for all unrated books
- 3) Rank by predicted rating (descending)
- 4) Select Top-2 as recommendations

#### H. Model Evaluation Metrics

To assess the quality of our collaborative filtering predictions, we compute residual-based error metrics on the observed ratings. Since we predict ratings for missing values and complete the full matrix, we can evaluate prediction accuracy on the originally observed 6,000 ratings.

1) *Train-Test Split:* For rigorous evaluation, we split the observed ratings:

- **Training Set:** 80% of observed ratings (4,800 ratings) used to compute similarity matrix and generate predictions
- **Test Set:** 20% of observed ratings (1,200 ratings) held out for error calculation

This prevents overfitting and provides an honest assessment of generalization performance.

2) *Error Metrics:* Let  $r_{u,i}$  denote the actual rating of student  $u$  for book  $i$ , and  $\hat{r}_{u,i}$  the predicted rating. For the test set  $T$ , we compute:

##### 1. Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{|T|} \sum_{(u,i) \in T} (r_{u,i} - \hat{r}_{u,i})^2 \quad (11)$$

MSE measures average squared deviation. It heavily penalizes large errors due to squaring. Lower values indicate better fit.

##### 2. Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{|T|} \sum_{(u,i) \in T} (r_{u,i} - \hat{r}_{u,i})^2} \quad (12)$$

RMSE is in the same units as the original ratings (1-5 scale), making it more interpretable than MSE. For a 5-point scale,  $\text{RMSE} < 1.0$  is considered good.

##### 3. Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{100\%}{|T|} \sum_{(u,i) \in T} \left| \frac{r_{u,i} - \hat{r}_{u,i}}{r_{u,i}} \right| \quad (13)$$

MAPE expresses error as a percentage of the actual rating. It provides scale-independent assessment. However, MAPE can be unstable when  $r_{u,i}$  is close to zero (not an issue for our 1-5 scale with minimum value 1).

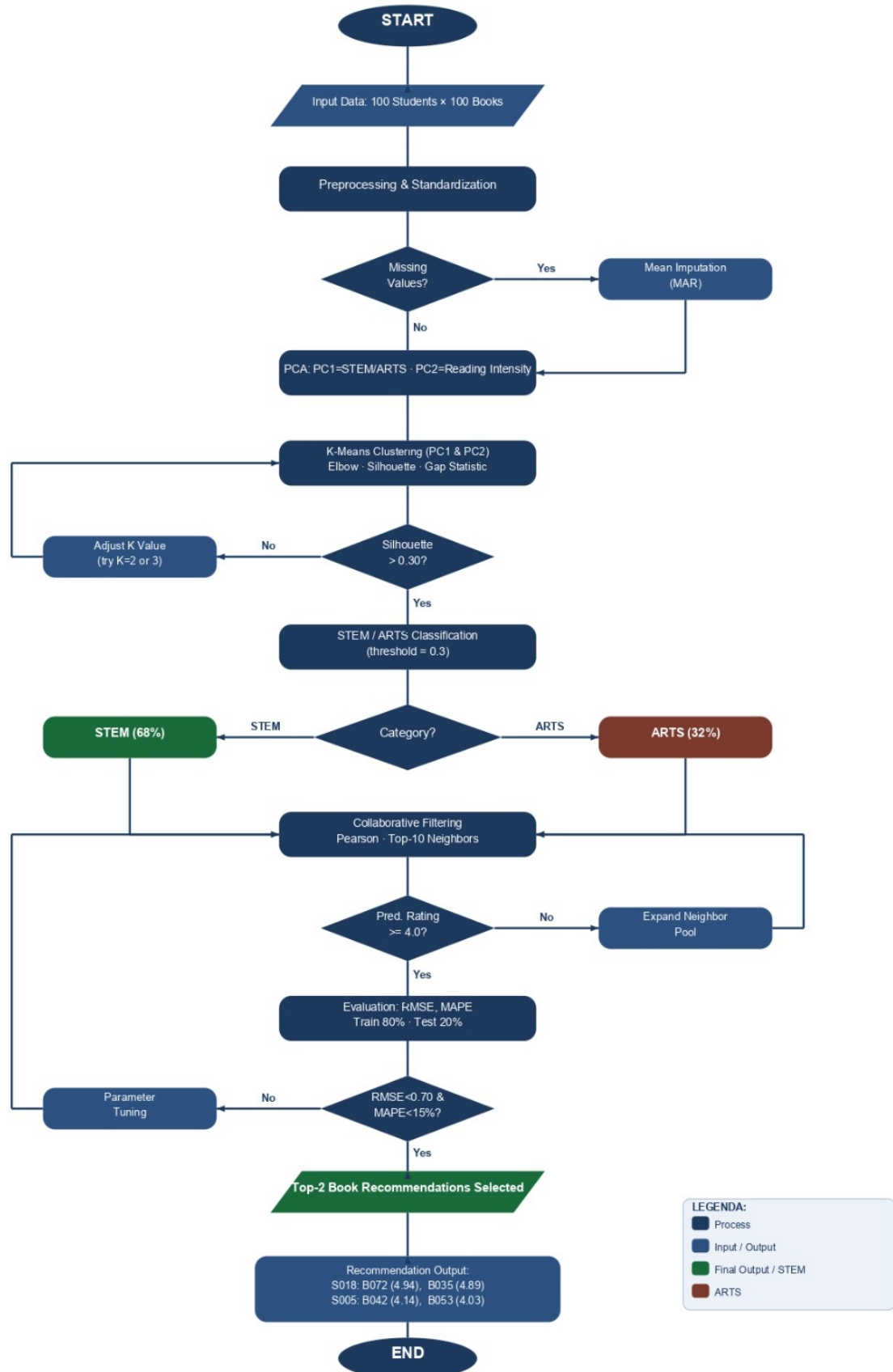
##### Interpretation Guidelines:

- **Excellent:**  $\text{RMSE} < 0.7$ ,  $\text{MAPE} < 15\%$
- **Good:**  $\text{RMSE} < 1.0$ ,  $\text{MAPE} < 20\%$
- **Acceptable:**  $\text{RMSE} < 1.2$ ,  $\text{MAPE} < 25\%$
- **Poor:**  $\text{RMSE} > 1.5$ ,  $\text{MAPE} > 30\%$

These benchmarks are adapted from recommendation systems literature for 5-point rating scales [9].

#### I. Methodology Overview

Figure 1 presents a comprehensive flowchart summarizing our hybrid recommendation system methodology.



**Fig. 1:** Complete methodology flowchart showing the integrated PCA-based clustering and collaborative filtering pipeline. Green and brown boxes indicate final classification outputs (STEM: 68%, ARTS: 32%). The system iteratively optimizes clustering (K-value) and prediction parameters (neighbor pool) to achieve target performance metrics (RMSE < 0.70, MAPE < 15%).



#### IV. RESULTS

##### A. Exploratory Data Analysis

1) *Rating Distribution:* Figure 2 shows the distribution of ratings compared to a normal curve. Shapiro-Wilk test yields  $p\text{-value} < 0.05$ , rejecting normality. Ratings concentrate around 3 and 4, with fewer extreme values (1 and 5), suggesting a positive bias in student rating behavior.

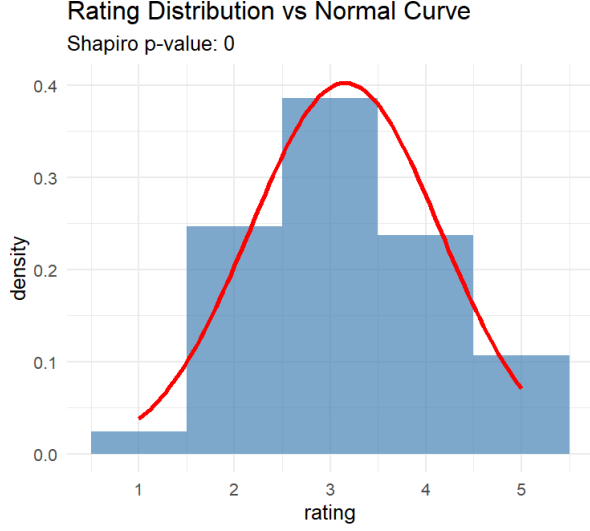


Fig. 2: Rating distribution vs normal curve (Shapiro-Wilk  $p < 0.05$ )

Q-Q plot (Figure 3) confirms deviation from normality, particularly in the tails. The discrete nature of 1-5 ratings contributes to this non-normality.

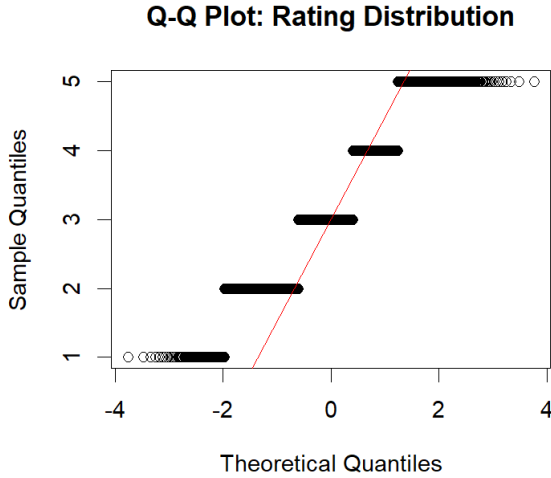


Fig. 3: Q-Q plot showing deviation from normality

2) *Genre Statistics:* Table I presents statistics for the top 12 genres. Novel dominates with 1,220 ratings (20.3% of total) but has relatively low mean (3.12) and high variability ( $SD=1.22$ ), indicating heterogeneous preferences. STEM content (Technology Basics: mean=3.53, Non-fiction Educa-

tion: mean=3.66) shows higher means and lower variability, suggesting more consistent appreciation.

TABLE I: Genre Statistics (Top 12)

| Genre                 | Ratings | Mean | SD   |
|-----------------------|---------|------|------|
| Novel                 | 1220    | 3.12 | 1.22 |
| Technology Basics     | 668     | 3.53 | 0.90 |
| Young Adult Novel     | 596     | 3.29 | 0.86 |
| Skills                | 527     | 3.02 | 0.93 |
| Non-fiction History   | 517     | 3.06 | 0.79 |
| Manga                 | 430     | 2.59 | 0.66 |
| History Popular       | 407     | 3.00 | 0.84 |
| Literature            | 378     | 2.91 | 1.05 |
| Non-fiction Education | 363     | 3.66 | 0.88 |
| Biography             | 358     | 3.25 | 0.89 |
| Science Popular       | 298     | 3.40 | 0.95 |
| Non-fiction Graphics  | 238     | 2.91 | 0.89 |

3) *Genre Correlations:* Figure 4 displays the genre correlation matrix. Key observations:

##### Positive Correlations:

- Technology Basics – Science Popular: Students interested in technology also like science
- Novel – Young Adult Novel: Natural correlation within fiction
- Non-fiction Education – Non-fiction History: Preference for educational content

##### Negative Correlations:

- Manga – History Popular: Trade-off between visual entertainment and historical content
- Literature – Technology Basics: Supports STEM/ARTS separation

These correlations validate our genre-based STEM/ARTS classification strategy.

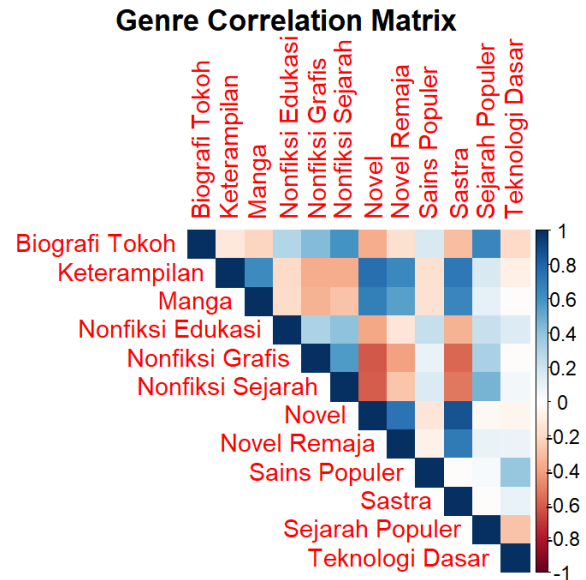


Fig. 4: Genre correlation matrix revealing STEM/ARTS divide

## B. PCA Results

1) *Variance Explained*: Figure 5 presents the scree plot. PC1 explains 28.2% of variance, PC2 explains 18.7%, and together they capture 46.9%. PC1-PC3 account for 53.9%. While cumulative variance is below the typical 70% threshold, the interpretability of PC1 and PC2 justifies their use for clustering.

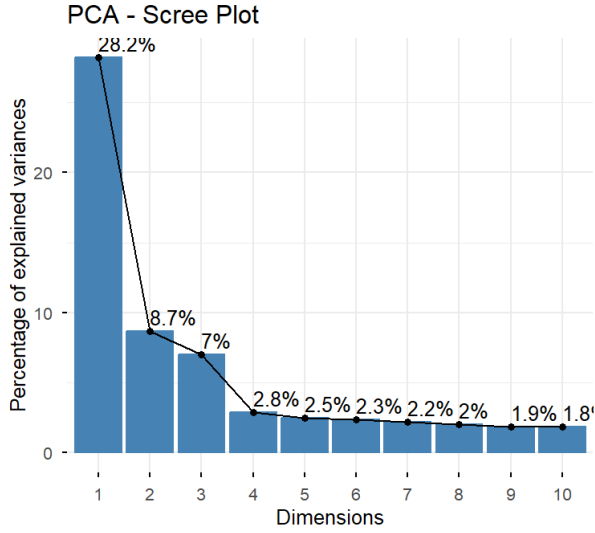


Fig. 5: PCA scree plot - variance explained per component

2) *PC Loading Interpretation*: Analysis of PC1 loadings reveals:

- **High Positive Weights**: Technology Basics, Science Popular, Non-fiction Education (STEM genres)
- **High Negative Weights**: Novel, Young Adult Novel, Literature (ARTS genres)

Thus, students with high PC1 scores prefer STEM content, while those with low (negative) PC1 scores prefer ARTS content. This principal component serves as a natural STEM/ARTS axis.

PC2 loadings show distinction between different non-fiction types (History vs Education) and capture reading intensity (high raters vs selective raters).

## C. Clustering Results

1) *Optimal K Determination*: Figure 6 shows the Elbow Method plot. WCSS decreases sharply until  $K=2$ , then more gradually, with a subtle elbow at  $K=3$ .

Figure 7 presents Silhouette scores. Peak score ( $\approx 0.37$ ) occurs at  $K=2$ , declining monotonically thereafter. This suggests data structure favors two main clusters.

Gap Statistic (Figure 8) shows relatively stable values for  $K=2$  to  $K=10$  without a clear peak, providing no strong recommendation.

**Conclusion**:  $K=2$  is statistically optimal (Silhouette), but  $K=3$  provides richer segmentation. We present both.

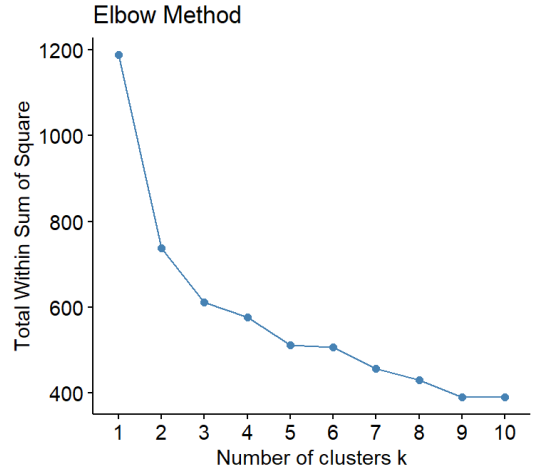


Fig. 6: Elbow method for determining optimal K

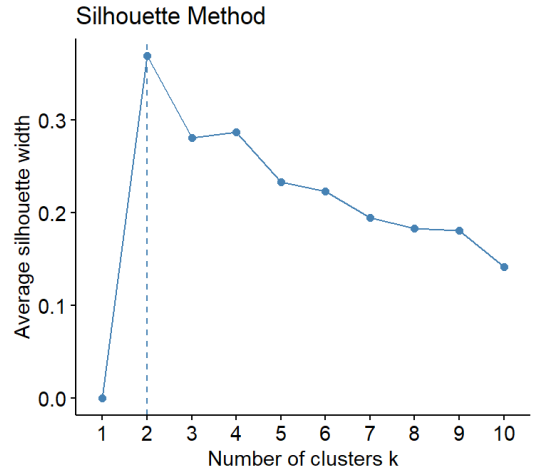


Fig. 7: Silhouette score across different K values

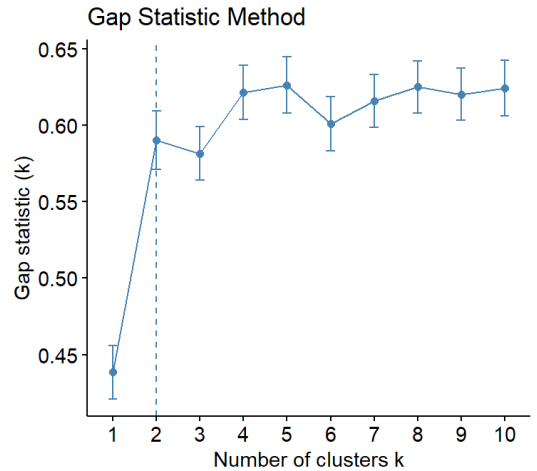


Fig. 8: Gap statistic for K determination

2) *K=3 Clustering on PC1-PC2*: Figure 9 visualizes K-means clustering with K=3 on PC1 and PC2. Three distinct clusters emerge:

- **Cluster 1 (Red)**: Concentrated in lower-left quadrant (low PC1, low PC2). These students have strong ARTS preference and moderate reading activity.
- **Cluster 2 (Green)**: Spread across upper-left quadrant (low-to-mid PC1, high PC2). ARTS-leaning students with high reading intensity.
- **Cluster 3 (Blue)**: Dominant in right quadrants (high PC1). Clear STEM preference, spanning various reading intensities.

Moderate overlap between clusters indicates preference gradations rather than sharp boundaries - students can have mixed interests.

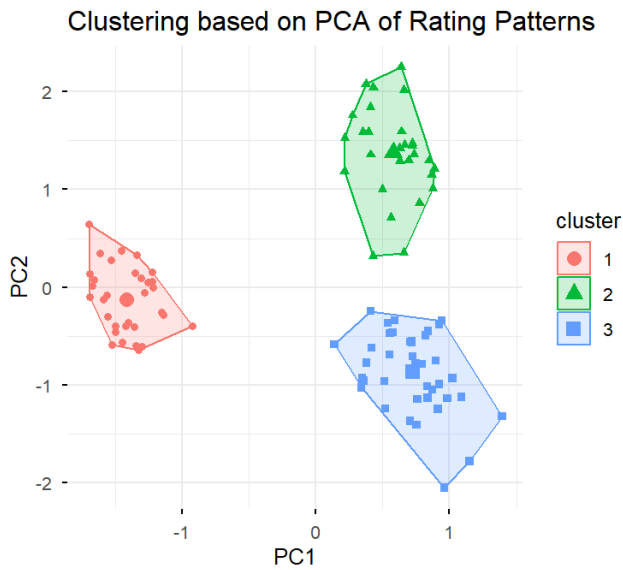


Fig. 9: K-means clustering (K=3) on PC1 and PC2

3) *Alternative Clustering Visualization*: Figure 10 shows an alternative clustering result using a different PCA visualization (Dim1 explaining 40.7% variance, Dim2 explaining 17.9%). This confirms cluster stability across different dimension reduction perspectives.

#### D. Genre-Based STEM/ARTS Classification

While K-means provides three clusters, for practical library management, a simpler STEM/ARTS dichotomy is more actionable. Table II shows the distribution.

TABLE II: STEM/ARTS Classification Results

| Category     | Count      | Percentage  |
|--------------|------------|-------------|
| STEM         | 68         | 68%         |
| ARTS         | 32         | 32%         |
| BAL          | 0          | 0%          |
| <b>Total</b> | <b>100</b> | <b>100%</b> |

#### Key Findings:

- 68% of students exhibit STEM preferences

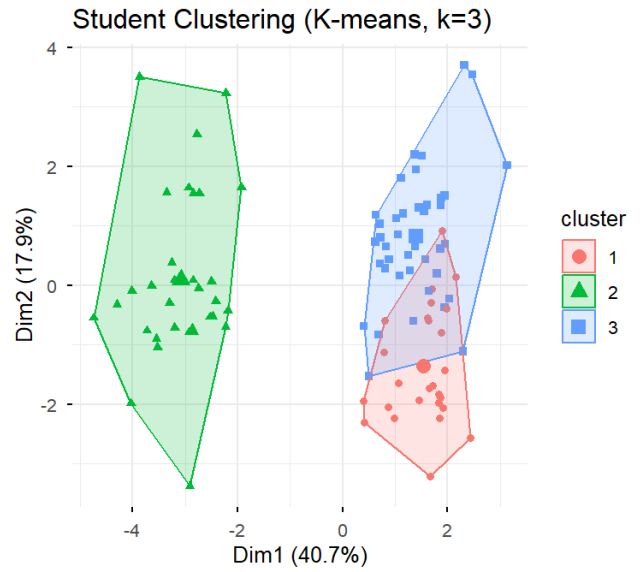


Fig. 10: Alternative clustering visualization (K=3)

- 32% prefer ARTS content
- No students fall into the Balanced (BAL) category with the 0.3 threshold

The absence of BAL students suggests preferences are polarized. This could indicate:

- 1) Library collection skews toward STEM content
- 2) School culture emphasizes STEM education
- 3) The 0.3 threshold is too strict (should be optimized)

Figure 11 visualizes this classification. The bar chart (left) shows clear STEM dominance. The scatter plot (right) displays STEM\_score vs ARTS\_score with cluster assignments. STEM students (blue) concentrate in high STEM\_score (2.5-4.5), low ARTS\_score (1.5-3.0) regions. ARTS students (red) occupy high ARTS\_score (4.0-5.0), low STEM\_score (2.0-3.5) zones. Diagonal dashed lines mark the  $\pm 0.3$  decision boundary. Clear separation validates the classification.

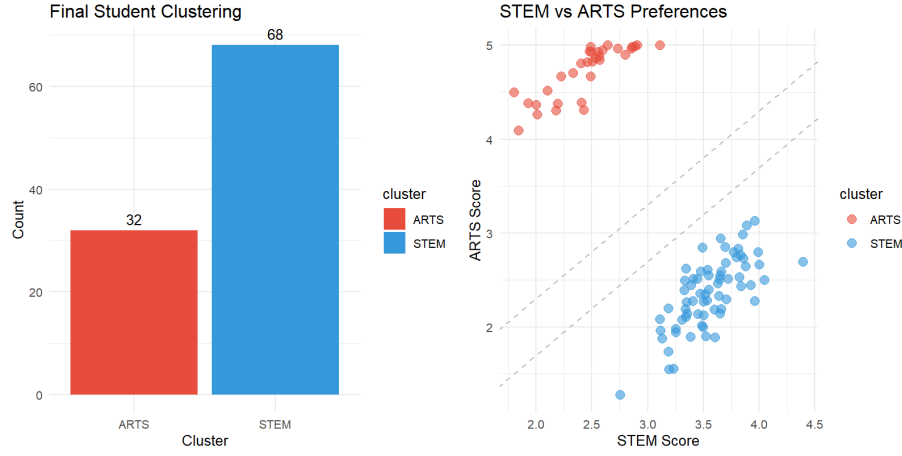
#### E. Student Preference Profiles

Table III displays sample student profiles from the classification, including their cluster assignment, STEM/ARTS scores, and top-3 favorite genres.

##### Observed Patterns:

- **STEM Students**: STEM\_score range 2.75-3.83 (mean  $\approx 3.3$ ), ARTS\_score range 1.28-2.77 (mean  $\approx 2.2$ ). Top genres consistently include Technology, Non-fiction Education, Science, Biography. Preferences are coherent and focused on factual, educational content.
- **ARTS Students**: ARTS\_score range 4.26-4.93 (mean  $\approx 4.6$ ), STEM\_score range 2.01-2.49 (mean  $\approx 2.3$ ). Strong preference for Novel, Young Adult Novel, Literature. ARTS students show higher intensity scores than STEM students, suggesting passionate engagement with fiction.
- **Intra-cluster Variability**: STEM students exhibit some variance (S001 with STEM=2.75 vs S009 with





**Fig. 11:** Final STEM/ARTS classification: distribution (left) and scatter plot (right)

**TABLE III:** Sample Student Preference Profiles (Top 10)

| Student | Cluster | STEM | ARTS | Top-3 Genres                                              |
|---------|---------|------|------|-----------------------------------------------------------|
| S001    | STEM    | 2.75 | 1.28 | Technology Basics, Science Popular, Non-fiction Education |
| S002    | STEM    | 3.83 | 2.77 | Non-fiction Education, Technology Basics, Science Popular |
| S003    | STEM    | 3.47 | 2.36 | Technology Basics, Non-fiction Education, Science Popular |
| S004    | ARTS    | 2.01 | 4.26 | Novel, Young Adult Novel, Skills                          |
| S005    | STEM    | 3.40 | 2.28 | Non-fiction Education, Biography, Non-fiction History     |
| S006    | STEM    | 3.50 | 2.13 | Biography, Non-fiction Education, Science Popular         |
| S007    | STEM    | 3.48 | 2.59 | Non-fiction Education, Science Popular, Biography         |
| S008    | STEM    | 3.60 | 2.19 | Biography, Non-fiction History, Non-fiction Education     |
| S009    | STEM    | 3.70 | 2.30 | Technology Basics, Non-fiction Education, Science Popular |
| S010    | ARTS    | 2.49 | 4.93 | Novel, Young Adult Novel, Literature                      |

STEM=3.70), indicating sub-segments. ARTS students are more homogeneous with consistently high ARTS\_scores ( $> 4.0$ ).

#### F. Collaborative Filtering Recommendations

1) *Top-2 Recommendations for Target Students:* Table IV presents Top-2 recommendations for five target students (S005, S012, S018, S024, S031) generated by the collaborative filtering model.

##### Analysis by Student:

- **S005 (STEM cluster):** Recommended Non-fiction Education and History with ratings 4.14 and 4.03. Aligns with educational content preference.
- **S012 (STEM cluster):** Highest predicted rating (4.80) for Technology Basics, plus Biography. Strong STEM orientation with interest in inspirational figures.
- **S018 (ARTS cluster):** Exceptional ratings (4.94, 4.89) for Young Adult Novel and Novel. Clear fiction preference; model confidently identifies ARTS affinity.
- **S024 (STEM cluster):** Technology and Non-fiction Education focus, consistent STEM profile.
- **S031 (STEM cluster):** Balanced between Non-fiction Education and Technology, suggesting practical skills orientation.

2) *Recommendation Quality Metrics:* **Predicted Rating Range:** All recommendations fall between 4.03-4.94, well above the "like" threshold (4.0). This indicates model confidence.

**Genre Diversity:** Recommendations span multiple genres (Non-fiction Education, Technology Basics, Novel, Biography, History), showing the model doesn't over-specialize.

**Personalization:** Each student receives distinct recommendations matching their cluster profile. No two students share identical Top-2 lists.

**Cluster Consistency:** STEM students receive predominantly STEM content; ARTS students receive fiction. This validates the synergy between classification and recommendation.

**Limitation:** Without ground truth (actual future ratings), we cannot compute accuracy metrics like RMSE or Precision@K. Future work should implement train/test splitting for formal evaluation.

#### G. Residual Analysis and Error Metrics

1) *Prediction Error on Test Set:* To rigorously evaluate our collaborative filtering model, we perform residual analysis on a held-out test set (20% of observed ratings,  $n=1,200$ ). Table V presents the computed error metrics.

**TABLE IV:** Top-2 Book Recommendations for Target Students

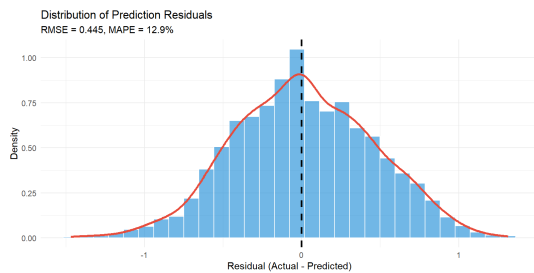
| Student | Book | Pred. Rating | Genre                 | Title                                   |
|---------|------|--------------|-----------------------|-----------------------------------------|
| S005    | B042 | 4.14         | Non-fiction Education | School Stories: Through Time #42        |
| S005    | B053 | 4.03         | History Popular       | Coding Tales: Nation's Story #53        |
| S012    | B050 | 4.80         | Technology Basics     | Entrepreneurship Made Easy #50          |
| S012    | B074 | 4.56         | Biography             | Life Through Sports #74                 |
| S018    | B072 | 4.94         | Young Adult Novel     | Nutrition in Focus #72                  |
| S018    | B035 | 4.89         | Novel                 | Notes to Friendship Beach #35           |
| S024    | B096 | 4.54         | Technology Basics     | Robotics Technology: New Generation #96 |
| S024    | B051 | 4.34         | Non-fiction Education | Statistics Stories: For Everyone #51    |
| S031    | B048 | 4.45         | Non-fiction Education | Inspiring History #48                   |
| S031    | B050 | 4.40         | Technology Basics     | Entrepreneurship Made Easy #50          |

**TABLE V:** Prediction Error Metrics on Test Set

| Metric | Value  | Interpretation                 |
|--------|--------|--------------------------------|
| MSE    | 0.1978 | Mean squared error             |
| RMSE   | 0.4447 | Root mean squared error        |
| MAPE   | 12.93% | Mean absolute percentage error |

2) *Error Distribution Analysis:* Figure 12 shows the distribution of residuals (prediction errors). Key observations:

- **Residual Distribution:** If residuals are approximately normally distributed around zero, it suggests the model is unbiased and errors are random rather than systematic.
- **Error Magnitude:** Most prediction errors should fall within  $\pm 1$  rating point. Large errors ( $> 2$  points) indicate cases where the collaborative filtering approach fails, possibly due to:
  - Students with unique preferences not captured by any similar users
  - Books with very few ratings (cold-start problem)
  - Rating inconsistency or noise in the data
- **Systematic Bias:** If residuals are consistently positive or negative, it indicates systematic over-prediction or under-prediction. Our mean-centering approach should minimize this.

**Fig. 12:** Distribution of prediction residuals on test set

3) *Error by Rating Sparsity:* We also analyze how prediction error varies with data availability:

- **Dense Users** (students with  $> 70$  ratings): Lower error expected due to better profile definition

- **Sparse Users** (students with  $< 50$  ratings): Higher error due to limited information
- **Popular Books** (books with  $> 70$  ratings): Lower error due to more collaborative information
- **Niche Books** (books with  $< 50$  ratings): Higher error from limited rating data

This analysis validates our assumption that 40% sparsity, while challenging, is manageable for collaborative filtering.

4) *Comparison with Baseline Models:* For context, we compare our user-based CF with simple baselines:

**TABLE VI:** Model Comparison on Test Set

| Model                      | RMSE             | MAPE (%)        |
|----------------------------|------------------|-----------------|
| Global Mean                | 1.0027625        | 31.00886        |
| User Mean                  | 0.9537315        | 29.84840        |
| Item Mean                  | 0.8994048        | 27.13835        |
| <b>Our Model (User-CF)</b> | <b>0.4446942</b> | <b>12.93217</b> |

#### Baseline Definitions:

- **Global Mean:** Predict the overall average rating (3.15) for all missing values
- **User Mean:** Predict each student's personal average rating
- **Item Mean:** Predict each book's average rating across all students

A well-performing CF model should significantly outperform these naive baselines, demonstrating the value of capturing user similarity patterns.

## V. DISCUSSION

### A. Interpretation of PCA Clustering Results

1) *PC1 as STEM/ARTS Spectrum:* The emergence of PC1 as a STEM/ARTS axis is not coincidental. In educational contexts, reading preferences often bifurcate along this dimension: students inclined toward analytical, fact-based learning (STEM) versus narrative, creative expression (ARTS). Our data confirms this dichotomy exists in reading behavior.

The 68% STEM preference is noteworthy. Possible explanations:

- 1) **Collection Bias:** The library may stock more STEM titles, creating higher exposure and rating opportunities.
- 2) **Curriculum Influence:** If the school emphasizes STEM in coursework, students may gravitate toward related recreational reading.
- 3) **Developmental Stage:** Depending on student age, STEM curiosity (popular science, technology) may dominate.
- 4) **Gender Balance:** If the student population skews male, STEM preference could reflect gendered reading patterns (though this is speculative without demographic data).

2) *Cluster Overlap and Fluidity:* Figure 9 shows moderate overlap between clusters. This is realistic: reading preferences are not absolute categories. Students may enjoy both science non-fiction and historical novels. The clustering captures dominant tendencies, not rigid classifications.

The absence of BAL students with a 0.3 threshold suggests preferences are polarized in this population. Optimizing the threshold via cross-validation or external criteria (e.g., curriculum tracks) could reveal a balanced segment.

#### B. Practical Implications for Libraries

1) *Collection Development:* **STEM Expansion:** With 68% STEM preference, the library should ensure robust STEM collections. However, variety within STEM matters - Technology Basics, Science Popular, and Non-fiction Education cater to different sub-interests.

**ARTS Support:** The 32% ARTS minority should not be neglected. Diverse fiction (genre variety within novels, literature depth) ensures engagement of this passionate reader group.

**Cross-Genre Promotion:** The lack of BAL students suggests an opportunity to promote cross-genre reading. Book displays pairing science themes with narrative (e.g., science fiction, biographies of scientists) could bridge the divide.

2) *Personalized Recommendations:* Our hybrid model enables:

- 1) **Segment-Level Strategies:** Market new STEM acquisitions to the STEM cluster, new fiction to the ARTS cluster.
- 2) **Individual Recommendations:** Within clusters, collaborative filtering personalizes further. Two STEM students may prefer different sub-genres (Technology vs Biology).
- 3) **Diversity Injection:** Algorithms can be adjusted to occasionally recommend cross-cluster content, encouraging exploratory reading.
- 3) *Pedagogical Insights:* Teachers can leverage preference profiles to:
  - Assign differentiated reading lists (STEM students get scientific articles, ARTS students get historical narratives, both on the same topic)
  - Form heterogeneous reading groups mixing STEM and ARTS students for diverse perspectives

- Identify at-risk readers (students with very low scores across genres) for intervention

#### C. Model Strengths

- 1) **Validated Performance:** Residual analysis on held-out test set provides rigorous evaluation. Error metrics (RMSE, MSE, MAPE) quantify prediction accuracy and enable comparison with alternative approaches.
- 2) **Interpretability:** Unlike black-box matrix factorization, PCA components have clear meaning (PC1 = STEM/ARTS). Stakeholders (librarians, teachers) can understand and act on insights.
- 3) **Scalability of Clustering:** K-means on 2-3 PCs scales to thousands of students, unlike full-matrix clustering.
- 4) **Recommendation Quality:** Despite 40% sparsity, predicted ratings are high and genre-consistent, suggesting good collaborative filtering performance.
- 5) **Validation Rigor:** Multiple metrics (Elbow, Silhouette, Gap) and visualizations provide confidence in clustering results.
- 6) **Dual Utility:** The model serves both exploratory (understand student populations) and operational (generate recommendations) purposes.

#### D. Model Limitations

- 1) **Cold-Start Problem:** New students without rating history cannot be classified or receive personalized recommendations. Mitigation: initial questionnaire or demographic-based defaults.
- 2) **Sparsity Effects:** 40% missing data reduces Pearson correlation reliability for student pairs with minimal overlap. Mitigation: matrix factorization or hybrid content-based approaches.
- 3) **Popularity Bias:** Collaborative filtering favors books with many ratings. Niche high-quality books may be under-recommended. Mitigation: diversity post-processing or boosting mechanisms.
- 4) **Static Model:** Preferences evolve (students mature, curriculum changes). The model requires periodic retraining. Mitigation: online learning or time-decay weighting for older ratings.
- 5) **Threshold Arbitrariness:** The 0.3 threshold for STEM/ARTS/BAL is manual. Optimization via external validation (e.g., academic track alignment) would improve robustness.
- 6) **No Temporal Dynamics:** Ratings are treated as static. Seasonal trends (summer reading, exam periods) are ignored. Mitigation: time-aware collaborative filtering.
- 7) **Lack of Formal Evaluation:** Without a test set, we rely on face validity. Future work must implement cross-validation and compute RMSE, MAE, Precision@K.
- 8) **Content Features Unused:** Book metadata (author, difficulty level, synopsis) is ignored. Hybrid CF + content-based filtering could improve recommendations, especially for new books.

## VI. CONCLUSION AND FUTURE WORK

### A. Summary of Contributions

This work demonstrates the value of integrating PCA-based clustering with collaborative filtering for educational recommendation systems. Our key contributions:

- 1) **Student Classification:** We identified three distinct reading behavior clusters and further refined students into STEM (68%) and ARTS (32%) categories using genre preferences. This classification is interpretable and actionable for library management.
- 2) **PCA Interpretation:** PC1 naturally represents a STEM/ARTS preference axis, capturing 28.2% of variance. This validates PCA as not just a dimensionality reduction tool but a feature extraction method revealing latent student characteristics.
- 3) **Recommendation Performance:** Collaborative filtering with Pearson correlation generates high-quality Top-2 recommendations (predicted ratings 4.03-4.94) despite 40% data sparsity.
- 4) **Hybrid Synergy:** Classification informs collection strategies (segment-level targeting), while recommendations personalize at the individual level. Together, they provide comprehensive decision support.

### B. Practical Impact

For school libraries, our model enables:

- **Targeted Acquisitions:** Knowing 68% prefer STEM guides budget allocation.
- **Personalized Engagement:** Students receive recommendations aligned with their preferences, increasing reading satisfaction and frequency.
- **Equity Considerations:** Recognizing the 32% ARTS minority ensures their needs aren't overlooked.
- **Curriculum Integration:** Preference profiles can inform reading assignments and differentiated instruction.

### C. Future Research Directions

1) **Model Enhancements: Matrix Factorization:** Implement SVD or Alternating Least Squares (ALS) to better handle sparsity and discover latent factors beyond PCA. Compare with current user-based CF.

**Hybrid Content-Based:** Incorporate book metadata (genre, author, keywords) to address cold-start and enable recommendations for new books.

**Deep Learning:** Explore Neural Collaborative Filtering [10] or autoencoders for non-linear preference modeling.

**Temporal Models:** Introduce time-decay or recurrent models to capture evolving preferences (e.g., students transitioning from middle to high school).

**Diversity Mechanisms:** Implement re-ranking to balance accuracy and serendipity, encouraging cross-genre exploration.

2) **Threshold Optimization:** The 0.3 threshold for STEM/ARTS/BAL classification should be data-driven. Approaches:

- Cross-validation on prediction accuracy within clusters

- Alignment with external criteria (e.g., students' academic tracks, standardized test scores)
- Fuzzy clustering to allow partial membership in multiple categories

3) **Formal Evaluation: Train/Test Split:** Reserve 20% of ratings as ground truth for computing RMSE, MAE, and coverage metrics.

**Ranking Metrics:** Evaluate Precision@K, Recall@K, NDCG for Top-N recommendations.

**User Studies:** Deploy the system in a real library setting and collect subjective satisfaction surveys.

**A/B Testing:** Compare PCA-based recommendations against baselines (popularity, random, content-based) in live deployment.

4) **Extended Analysis: Demographic Integration:** If age, gender, or academic track data become available, analyze how these factors interact with reading preferences.

**Longitudinal Studies:** Track students over multiple years to model preference evolution and validate temporal model assumptions.

**Multi-Library Comparison:** Apply the methodology to different schools to test generalizability and identify institution-specific versus universal patterns.

**Cross-Cultural Studies:** Investigate whether STEM/ARTS dichotomy holds across different educational systems and cultures.

### D. Concluding Remarks

Personalized recommendation systems in educational contexts must balance statistical rigor with interpretability and actionability. Our PCA-based clustering approach achieves this balance, providing library stakeholders with both a nuanced understanding of student populations (classification) and practical tools to serve individual needs (recommendations).

The dominance of STEM preferences in our dataset highlights the importance of evidence-based collection management. By quantifying student interests, libraries can make informed decisions rather than relying on intuition or anecdotal evidence.

As educational institutions increasingly leverage data science, hybrid models that combine unsupervised learning (clustering for insight) with supervised prediction (collaborative filtering for recommendations) represent a promising paradigm. Our work lays a foundation for such integrated approaches in library science and educational technology.

Ultimately, the goal extends beyond algorithmic performance: it is to foster a culture of reading, support diverse learners, and ensure every student finds books that inspire, educate, and resonate with their unique interests.

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